

Coleridge Faraday

Education

- PhD in Physics, University of Cape Town** 2025–Present
○ Supervisor: Assoc. Prof. W. A. Horowitz
- MSc with Distinction in Physics, University of Cape Town (90%)** 2023–2024
○ Thesis: [Energy Loss and Theoretical Uncertainties in Small Quark-Gluon Plasmas](#)
○ Supervisor: Assoc. Prof. W. A. Horowitz
- Honours in Physics, University of Cape Town (85%)** 2022
○ Supervisor: *Investigating the short path length corrections to jet phenomenology in heavy ion collisions*
○ Supervisor: Assoc. Prof. W. A. Horowitz
- BSc in Physics and Mathematics, University of Cape Town (81%)** 2019–2021
○ Supervisor: *Spontaneous symmetry breaking and the Higgs mechanism.*
○ Supervisor: Assoc. Prof. Heribert Weigert

Awards

- Best PhD Oral Presentation in Theoretical and Computational Physics at SAIP2025 Annual Conference 2025
[National Institute for Theoretical and Computational Sciences \(NITHECS\) Scholarship](#) 2025 – 2027
Science Faculty PhD Fellowship, University of Cape Town 2025 – 2027
Manuel & Luby Washkansky Scholarship, University of Cape Town 2023
[SA-CERN Excellence Scholarship](#) 2023 – 2024
Science Faculty Scholarship, University of Cape Town 2020
Science Faculty Entrance Scholarship, University of Cape Town 2019
Dean's Merit List, University of Cape Town 2019, 2021, *
[UCT Mathematics Olympiad Gold Award](#) 2015, 2016
Professor John Webb Mathematics Scholarship, Wynberg Boys' High School 2014 – 2018

Publications

Journal Articles

- [1] **Faraday, C. & Horowitz, W. A. May 2025.** Statistical analysis of pQCD energy loss across system size, 2025 flavor, $\sqrt{s_{NN}}$, and p_T . *Submitted to JHEP*. arXiv: [2505.14568 \[hep-ph\]](#).
- [2] **Faraday, C. & Horowitz, W. A. May 2025.** Collisional and radiative energy loss in small systems. *Phys. Rev. C* **111**, 054911. doi:[10.1103/PhysRevC.111.054911](#). arXiv: [2408.14426 \[nucl-th\]](#).
- [3] **Faraday, C. & Horowitz, W. A. Mar. 2025.** A unified description of small, peripheral, and large system suppression data from pQCD. *Phys. Lett. B* **864**, 139437. doi:[10.1016/j.physletb.2025.139437](#). arXiv: [2411.09647 \[hep-ph\]](#).
- [4] **Faraday, C., Grindrod, A. & Horowitz, W. A. Nov. 2023.** Inconsistencies in, and short pathlength 2023 correction to, $R_{AA}(p_T)$ in A + A and p + A collisions. *Eur. Phys. J. C* **83**, 1060. doi:[10.1140/epjc/s10052-023-12234-y](#). arXiv: [2305.13182 \[hep-ph\]](#).

Conference Papers

- [1] **Faraday, C. & Horowitz, W. A. Jan. 2025.** *Heavy Flavour Energy Loss in Small and Large Systems* in. **316.** 2025 Contribution to SQM2024 (Jan.), 04001. doi:[10.1051/epjconf/202531604001](#). arXiv: [2409.14886 \[nucl-th\]](#).
- [2] Bert, B., **Faraday, C. & Horowitz, W. A. Dec. 2024.** *Modeling the Geometry of the Quark Gluon Plasma* 2024 in *68th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2024 (Dec.), 618–623. ISBN: 978-1-0370-2645-4. <https://www.coleridgefaraday.com/assets/pdf/papers/saipBert2024.pdf>.
- [3] **Faraday, C. & Horowitz, W. A. Dec. 2024.** *Energy Loss in Small Quark Gluon Plasmas* in *68th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2024 (Dec.), 629–634. ISBN: 978-1-0370-2645-4. <https://www.coleridgefaraday.com/assets/pdf/papers/saipFaraday2024.pdf>.
- [4] **Faraday, C. & Horowitz, W. A. June 2024.** *High- p_T suppression in small systems* in. **296.** Contribution to Quark Matter 2023 (June), 12011. doi:[10.1051/epjconf/202429612011](#). arXiv: [2312.10712 \[hep-ph\]](#).
- [5] **Faraday, C. & Horowitz, W. Feb. 2024.** *Energy Loss in Small Collision Systems* in. **HardProbes2023.** Contribution to Hard Probes 2023 (Feb.), 131. doi:[10.22323/1.438.0131](#). arXiv: [2307.08355 \[nucl-th\]](#).
- [6] **Faraday, C. & Horowitz, W. A. Sept. 2023.** *Assumption Breakdown in Radiative Energy Loss* in *67th Annual Conference of the South African Institute of Physics* Contribution to SAIP 2023 (Sept.). arXiv: [2309.06246 \[hep-ph\]](#).

Theses

- [1] **Faraday, C. Dec. 2024.** *Energy loss and theoretical uncertainties in small quark-gluon plasmas* Other 2024 thesis (Dec.). arXiv: [2506.02056 \[hep-ph\]](#).

Talks and Posters

Slides for my major presentations are available [here](#) under a CC-BY license.

*No awards were given in 2020 due to the COVID-19 pandemic

Conference Talks

- [1] **Faraday, C. Sept. 2025.** *Statistical Analysis of Energy Loss Across System Size* Precision Frontier of QCD Matter: Inference and Uncertainty Quantification. Central China Center for Nuclear Theory, Central China Normal University, Wuhan, China, Sept. **2025**
- [2] **Faraday, C. July 2025.** *Energy Loss as a Probe of Quark-Gluon Plasma Formation Across Collision System Size* 69th Annual Conference of the South African Institute of Physics (SAIP 2025). University of the Witwatersrand, Johannesburg, South Africa, July.
- [3] **Faraday, C. July 2024.** *Energy Loss in Small Quark Gluon Plasmas* 68th Annual Conference of the South African Institute of Physics (SAIP 2024). Makhanda, South Africa, July. **2024**
- [4] **Faraday, C. June 2024.** *Heavy Flavour Energy Loss in Small and Large Systems* The 21st International Conference on Strangeness in Quark Matter (SQM 2024). Strasbourg, France, June.
- [5] **Faraday, C. Sept. 2023.** *High- p_T suppression in small systems* 30th International Conference on Ultrarelativistic Nucleus-Nucleus Collisions (Quark Matter 2023). Houston, Texas, USA, Sept. **2023**
- [6] **Faraday, C. July 2023.** *Short path length corrections to pQCD phenomenological energy loss models in a QGP* 67th Annual Conference of the South African Institute of Physics (SAIP 2023). Richard's Bay, South Africa, July.

Conference Posters

- [1] **Faraday, C. Apr. 2025.** *Data-Driven Predictions for Small System Energy Loss* Quark Matter 2025. Goethe University Frankfurt: Quark Matter 2025, Apr. **2025**

Invited Talks

- [1] **Faraday, C. June 2025.** *Is there energy loss in small systems?* Set of three invited talks. New Mexico State University Nuclear Theory Group (Virtual), June.

Outreach Talks

- [1] Brand, J., **Faraday, C.** & Roosenstein, R. **July 2025.** *Outreach talk to disadvantaged primary school students* In collaboration with [Bartosch Foundation](#). Masiphumelele, South Africa, July.
- [2] **Faraday, C. May 2025.** *Smashing Atoms to Rewind the Universe* The University of Cape Town's Society of Physics Students. Cape Town, South Africa, May.
- [3] **Faraday, C. Oct. 2024.** *Racing Balls* Phenomenal physics (high school outreach). Cape Town, South Africa: University of Cape Town, Oct. **2024**
- [4] **Faraday, C. Sept. 2024.** *Introduction to Heavy-Ion Physics* Outreach trip to University of Venda. Thohoyandou, South Africa: University of Venda, Sept.
- [5] **Faraday, C. Nov. 2023.** *Quark Gluon Plasma Formation in Small Systems* ATLAS Showcase, UCT-CERN relaunch. Cape Town, South Africa, Nov. **2023**
- [6] **Faraday, C. Oct. 2023.** *Probing the Early Universe with Extreme Temperature Physics* Postgraduate Showcase 2023. Cape Town, South Africa: University of Cape Town, Oct.

Academic Service

Postgraduate representative for Physics Department, University of Cape Town

2025 – Present

Reviewing

Proceedings of the 68th Annual Conference of the South African Institute of Physics
Proceedings of the 69th Annual Conference of the South African Institute of Physics

Teaching

Quantum Field Theory II (SATACS), Tutor	2025
Quantum Field Theory I (SATACS), Tutor	2024,2025
Honours Statistical Physics (UCT), Tutor	2024
Honours Quantum Mechanics (UCT), Tutor	2023, 2024
Third Year Statistical Physics (UCT), Tutor	2023,2025
First Year Physics (with laboratory) (UCT), Tutor	2022
High School Physics, Mathematics, Chemistry, and Computer Science (Chit-Chat Tutoring), Private Tutor	2019–2022

Mentorship roles

Ben Bert. Mentoring and helping with honours and MSc projects on phenomenological energy loss at UCT	2023 – Present
Tiaan van der Merwe. Mentoring and helping with honours project on phenomenological energy loss at UCT	2024 – Present
Jack Brand. Mentoring and helping with honours project on phenomenological energy loss at UCT	2025 – Present